



Mortgage Pricer — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 26-March-2026

David K Kelly

CONFIDENTIAL — PROPRIETARY

Legal Notice

PROPRIETARY AND CONFIDENTIAL

This document, including all algorithms, models, methodology, formulas, calculations, and intellectual concepts contained herein, constitutes the exclusive intellectual property and confidential trade secrets of MKM Research Labs (“MKM”).

Any usage, reproduction, distribution, modification, or disclosure of this document or any portion thereof, without the express prior written authorisation from MKM Research Labs is strictly prohibited and constitutes an infringement of intellectual property rights.

The document is provided on an “as is” basis. MKM Research Labs makes no representations or warranties of any kind, express or implied, regarding its accuracy, reliability, or suitability for any particular purpose.

All rights reserved. © 2021–2026 MKM Research Labs.

Governed by the laws of the United Kingdom.

1 Test Results — Mortgage Pricer (MKM-MP-001)

Tests run on **26 March 2026 at 18:58:34**.

Table 1: Test Results Summary — Mortgage Pricer (MKM-MP-001)

Total Tests	67
Passed	67
Failed	0
Skipped	0
Duration	0.04s

Table 2: Detailed Test Results — Mortgage Pricer

Test Description	Acceptance Criteria	Result	Time
Default ids assigned	<code>results[0]["mortgage_id"] == "MORTGAGE_0";</code> <code>results[0]["property_id"] == "PROPERTY_0"</code>	Pass	0.001s
Error isolation	<code>len(results) == 3;</code> "error" not in <code>results[0]</code> (+2 more)	Pass	0.001s
Prices multiple mortgages	<code>len(results) == 2;</code> <code>results[0]["mortgage_id"] == "M1"</code> (+1 more)	Pass	0.001s
Custom tax rate	<code>MortgagePricer(tax_rate=0.40).tax_rate == 0.40</code>	Pass	0.000s
Default tax rate	<code>MortgagePricer().tax_rate == 0.20</code>	Pass	0.000s
Custom tax rate	<code>spread_high_tax > spread_low_tax</code>	Pass	0.000s
Floor at 10bps	<code>spread >= 0.001</code>	Pass	0.000s
Higher affordability higher spread	<code>spread_low_income > spread_high_income</code>	Pass	0.000s
Negative income fallback	<code>spread == 0.15</code>	Pass	0.000s
Spread bounded by schedule	<code>spread >= 0.001; spread < 0.05</code>	Pass	0.000s
Zero income fallback	<code>spread == 0.15</code>	Pass	0.000s
debug=True in the credit spread calculation.	<code>"credit_spread" in result</code>	Pass	0.001s
debug=True exercises logging branches without error.	<code>result["mortgage_value"] > 0</code>	Pass	0.001s
Zero income with debug=True exercises zero-income debug branch.	<code>result["credit_spread"] > 0</code>	Pass	0.001s
flood_risk_category=None with debug=True.	<code>result["mortgage_value"] > 0</code>	Pass	0.001s

Test Description	Acceptance Criteria	Result	Time
When interest_rate=0, monthly_rate==0 → loan_amount/n_periods formula.	result["mortgage_value"] > 0	Pass	0.001s
Batch passes flood risk	results[0]["flood_risk_factor"] == 1.40; results[1]["flood_risk_factor"] == 1.00 (+1 more)	Pass	0.001s
Case insensitive	MortgagePricer.calculate_flood_risk_impact("high") =...; MortgagePricer.calculate_flood_risk_impact("VERY LOW") ==... (+1 more)	Pass	0.000s
Flood factor in result	result["flood_risk_factor"] == 1.20	Pass	0.001s
Flood risk increases spread	high["credit_spread"] > low["credit_spread"]; high["flood_risk_factor"] > low["flood_risk_factor"] (+1 more)	Pass	0.001s
Flood risk multipliers[high-1.4] [category='High', expected=1.4]	MortgagePricer.calculate_flood_risk_impact(category) == e...	Pass	0.000s
Flood risk multipliers[low-1.05] [category='Low', expected=1.05]	MortgagePricer.calculate_flood_risk_impact(category) == e...	Pass	0.000s
Flood risk multipliers[medium-1.2] [category='Medium', expected=1.2]	MortgagePricer.calculate_flood_risk_impact(category) == e...	Pass	0.000s
Flood risk multipliers[very high-1.75] [category='Very High', expected=1.75]	MortgagePricer.calculate_flood_risk_impact(category) == e...	Pass	0.000s
Flood risk multipliers[very low-1.0] [category='Very Low', expected=1.0]	MortgagePricer.calculate_flood_risk_impact(category) == e...	Pass	0.000s
No flood risk defaults to unity	result["flood_risk_factor"] == 1.0	Pass	0.001s
None returns unity	MortgagePricer.calculate_flood_risk_impact(None) == 1.0	Pass	0.000s
Unknown category returns unity	MortgagePricer.calculate_flood_risk_impact("Unknown") == 1.0; MortgagePricer.calculate_flood_risk_impact("Extreme") == 1.0	Pass	0.000s
Ltv thresholds[0.5-1.0] [ltv=0.5, expected=1.0]	pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...	Pass	0.000s
Ltv thresholds[0.8-1.0] [ltv=0.8, expected=1.0]	pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...	Pass	0.000s
Ltv thresholds[0.85-1.1] [ltv=0.85, expected=1.1]	pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Ltv thresholds[0.9-1.1] [ltv=0.9, expected=1.1]	<code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>	Pass	0.000s
Ltv thresholds[0.91-1.3] [ltv=0.91, expected=1.3]	<code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>	Pass	0.000s
Ltv thresholds[0.95-1.3] [ltv=0.95, expected=1.3]	<code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>	Pass	0.000s
Ltv thresholds[0.96-1.5] [ltv=0.96, expected=1.5]	<code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>	Pass	0.000s
Ltv thresholds[1.0-1.5] [ltv=1.0, expected=1.5]	<code>pricer.calculate_loan_to_value_impact(pv * ltv, pv) == ex...</code>	Pass	0.000s
Zero property value	<code>pricer.calculate_loan_to_value_impact(100.0, 0) == 1.5</code>	Pass	0.000s
Higher rate higher payment	<code>high > low</code>	Pass	0.000s
Shorter term higher payment	<code>short > long</code>	Pass	0.000s
£200k at 3.5% over 25yr £1,001.25/month.	<code>995 < payment < 1010</code>	Pass	0.000s
Zero-rate degenerate case: $M = P / n$.	<code>payment == pytest.approx(1000.0)</code>	Pass	0.000s
Each mortgage has property id	<code>mapping.get("property_id") is not None</code>	Pass	0.002s
Processing stats	<code>stats["successful_mortgages"] == 5;</code> <code>stats["failed_mortgages"] == 0</code>	Pass	0.002s
Output file exists	<code>result["file_path"].exists()</code>	Pass	0.002s
Output file is valid json	<code>isinstance(data, dict)</code>	Pass	0.004s
Generate returns expected keys	<code>"data" in result;</code> <code>"file_path" in result (+1 more)</code>	Pass	0.002s
Mortgage count matches properties	<code>len(result["data"]["mortgages"]) == 5</code>	Pass	0.002s
Mortgage id format	<code>re.match(r"~MORT-[a-f0-9]{8}\$", mid)</code>	Pass	0.002s
Mortgage ids unique	<code>len(ids) == len(set(ids))</code>	Pass	0.003s
All errors returns error	<code>"error" in calculate_portfolio_metrics(results)</code>	Pass	0.000s
High risk count	<code>calculate_portfolio_metrics(results)["high_risk_mortgages"]</code>	Pass	0.000s
Mixed results exclude errors	<code>metrics["total_mortgages"] == 1;</code> <code>metrics["error_count"] == 1</code>	Pass	0.001s
Valid results	<code>metrics["total_mortgages"] == 2;</code> <code>metrics["error_count"] == 0 (+3 more)</code>	Pass	0.001s
Balance converges to zero	<code>result["outstanding_balance"][-1] < 1.0</code>	Pass	0.001s
Expected losses non negative	<code>all(e1 >= 0 for e1 in result["expected_losses"])</code>	Pass	0.001s

Test Description	Acceptance Criteria	Result	Time
Fair value less than par	<code>result["mortgage_value"] < base.mortgage_params["loan_amo...; result["discount_to_par"] > 0 (+1 more)</code>	Pass	0.001s
Hazard rates positive	<code>result["hazard_rates"][0] == 0.0; all(result["hazard_rates"][i] > 0 for i in range(1, len(r...</code>	Pass	0.001s
High ltv increases spread	<code>high_ltv["credit_spread"] > low_ltv["credit_spread"]; high_ltv["ltv_factor"] > low_ltv["ltv_factor"]</code>	Pass	0.001s
Higher haircut more loss	<code>high["pv_losses"] > low["pv_losses"]</code>	Pass	0.001s
Input validation clamps	<code>result["mortgage_value"] > 0</code>	Pass	0.000s
Ltv factor in result	<code>"ltv_factor" in result; result["ltv_factor"] == 1.0</code>	Pass	0.001s
Raises on invalid types	—	Pass	0.000s
Recovery haircut zero low ltv	<code>all(lgd == 0 for lgd in result["lgds"])</code>	Pass	0.001s
Survival first period less than one	<code>result["survival_probs"][1] < 1.0</code>	Pass	0.001s
Survival monotonically decreasing	<code>surv[i] <= surv[i - 1]</code>	Pass	0.001s
Positive rate exceeds principal	<code>cost > 200.000</code>	Pass	0.000s
Zero rate equals principal	<code>cost == pytest.approx(100_000.0)</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (65 tests)	D. Kelly
07-Mar-2026	1.1	62 test(s) added; 65 test(s) removed (total: 62)	D. Kelly
26-Mar-2026	1.2	5 test(s) added (total: 67)	D. Kelly